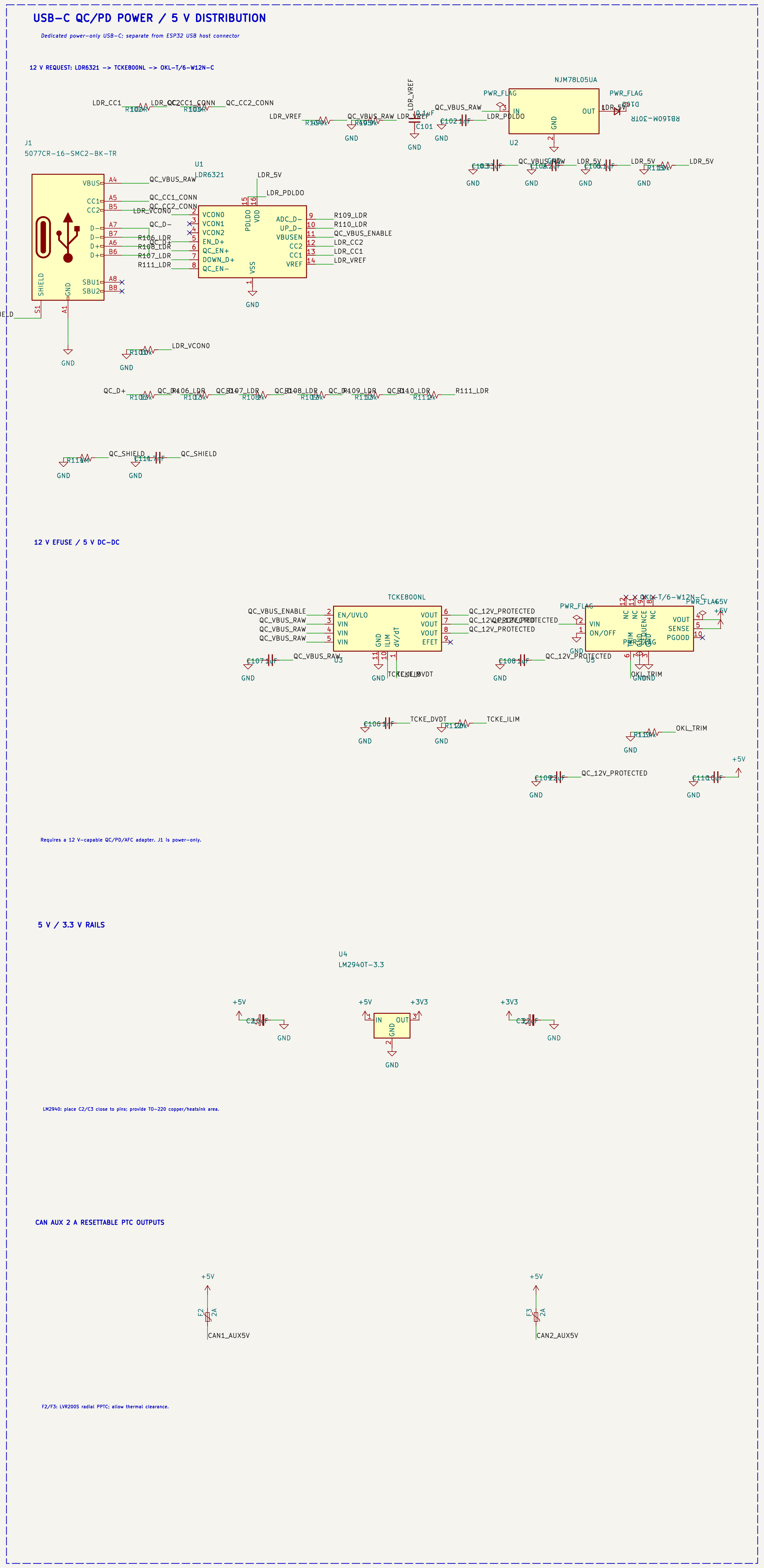
USB-C QC/PD POWER / 5 V DISTRIBUTION

Dedicated power-only USB-C; separate from ESP32 USB host connector

12 V REQUEST: LDR6321 -> TCKE800NL -> OKL-T/6-W12N-C


J1
5077CR-16-SMC2-BK-TR
U1
LDR6321
U2
NJM78L05UA
QC_VBUS_RAW
LDR_5V
LDR_VREF
LDR_CC1
QC_CC1_CONN
QC_CC2_CONN
QC_SHIELD
LDR_VCONN
PWR_FLAG
GND
12 V EFUSE / 5 V DC-DC
TCKE800NL
U3
QC_VBUS_ENABLE
QC_12V_PROTECTED
TCKE_DVDT
TCKE_ILIM
OKL-T/6-W12N-C
OKL_TRIM
+5V
5 V / 3.3 V RAILS
U4
LM2940T-3.3
+3V3
CAN AUX 2 A RESETTABLE PTC OUTPUTS
F2
2A
CAN1_AUX5V
CAN2_AUX5V

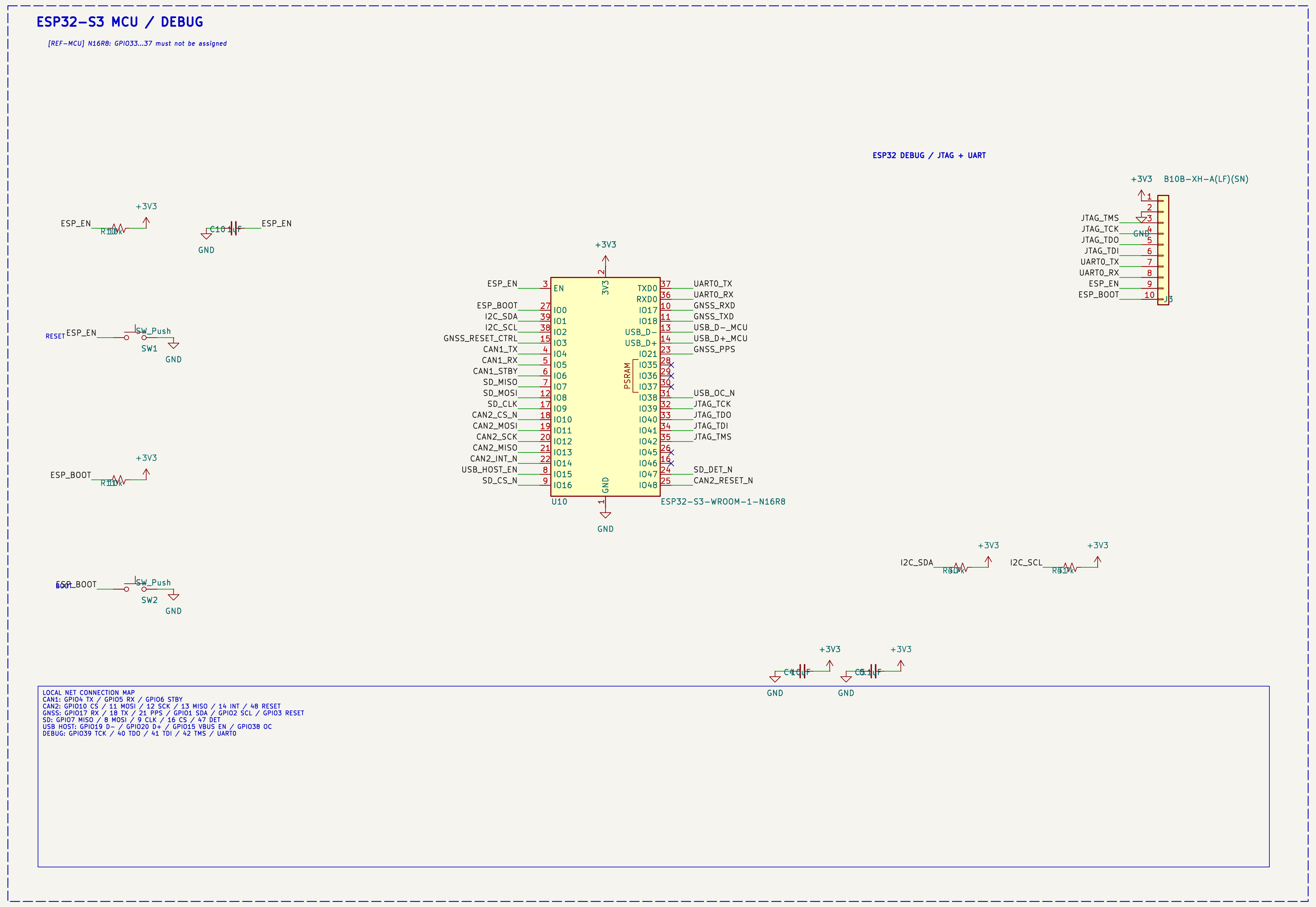
Requires a 12 V-capable QC/PD/AFC adapter. J1 is power-only.

LM2940: place C2/C3 close to pins; provide TO-220 copper/heatsink area.

F2/F3: LVR200S radial PPTC; allow thermal clearance.

ESP32-S3 MCU / DEBUG

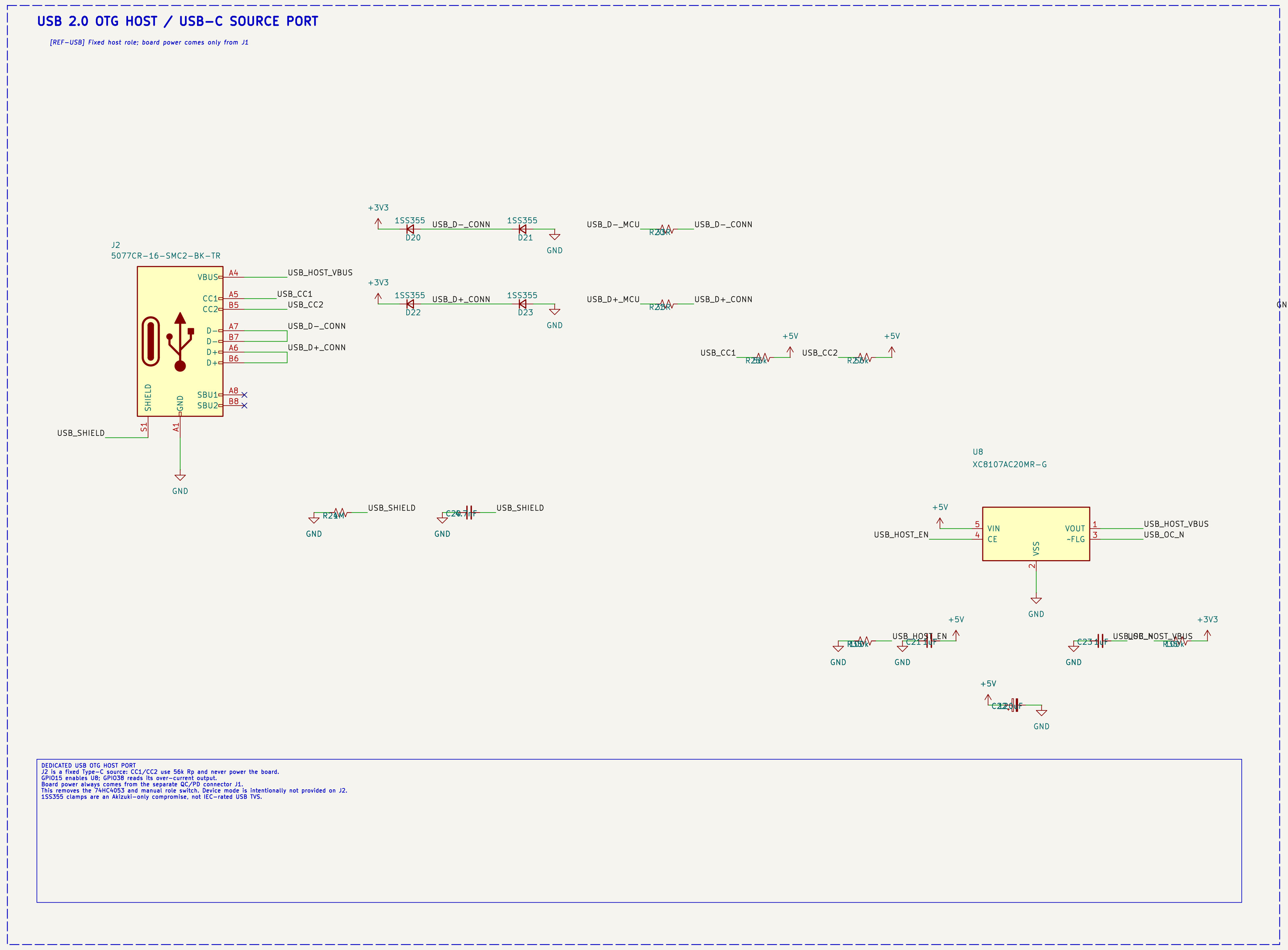
[REF-MCU] N16R8: GPIO33...37 must not be assigned


ESP32 DEBUG / JTAG + UART
U10
ESP32-S3-WROOM-1-N16R8
ESP_EN
ESP_BOOT
SW1
SW2
SW_Push
B10B-XH-A(LF)(SN)
JTAG_TMS
JTAG_TCK
JTAG_TDO
JTAG_TDI
UART0_TX
UART0_RX
I2C_SDA
I2C_SCL
GNSS_RESET_CTRL
CAN1_TX
CAN1_RX
CAN1_STBY
SD_MISO
SD_MOSI
SD_CLK
CAN2_CS_N
CAN2_MOSI
CAN2_SCK
CAN2_MISO
CAN2_INT_N
USB_HOST_EN
SD_CS_N
GNSS_RXD
GNSS_TXD
USB_D-_MCU
USB_D+_MCU
GNSS_PPS
USB_OC_N
SD_DET_N
CAN2_RESET_N
+3V3
GND

LOCAL NET CONNECTION MAP
CAN1: GPIO4 TX / GPIO5 RX / GPIO6 STBY
CAN2: GPIO10 CS / 11 MOSI / 12 SCK / 13 MISO / 14 INT / 48 RESET
GNSS: GPIO17 RX / 18 TX / 21 PPS / GPIO1 SDA / GPIO2 SCL / GPIO3 RESET
SD: GPIO7 MISO / 8 MOSI / 9 CLK / 16 CS / 47 DET
USB HOST: GPIO19 D- / GPIO20 D+ / GPIO15 VBUS EN / GPIO38 OC
DEBUG: GPIO39 TCK / 40 TDO / 41 TDI / 42 TMS / UART0

USB 2.0 OTG HOST / USB-C SOURCE PORT

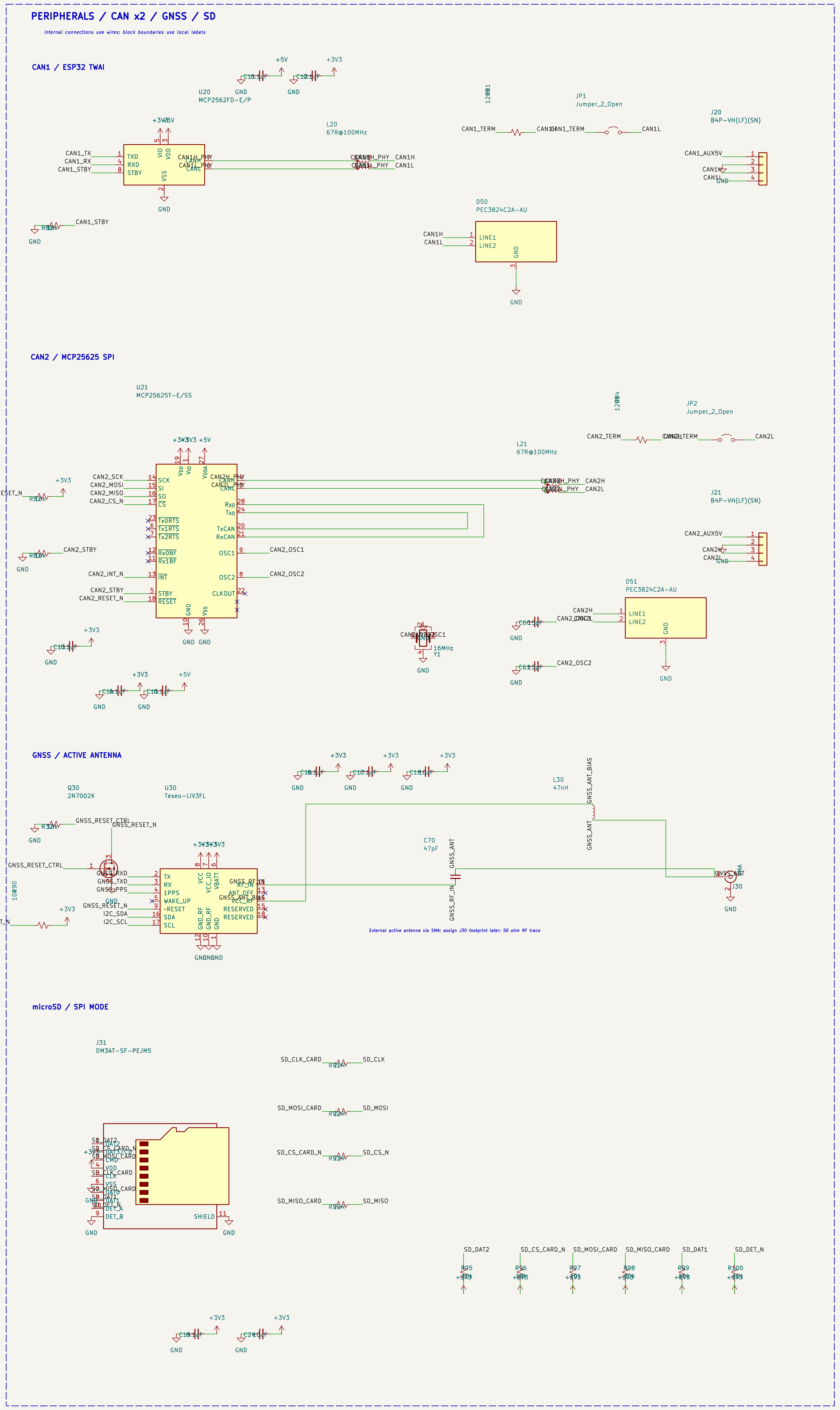
[REF-USB] Fixed host role; board power comes only from J1


J2
5077CR-16-SMC2-BK-TR
USB_HOST_VBUS
USB_CC1
USB_CC2
USB_D-_CONN
USB_D+_CONN
USB_SHIELD
1SS355
D20
D21
D22
D23
USB_D-_MCU
USB_D+_MCU
U8
XC8107AC20MR-G
USB_HOST_EN
USB_OC_N
+5V
+3V3
GND

DEDICATED USB OTG HOST PORT
J2 is a fixed Type-C source: CC1/CC2 use 56k Rp and never power the board.
GPIO15 enables U8; GPIO38 reads its over-current output.
Board power always comes from the separate QC/PD connector J1.
This removes the 74HC4053 and manual role switch. Device mode is intentionally not provided on J2.
1SS355 clamps are an Akizuki-only compromise, not IEC-rated USB TVS.

PERIPHERALS / CAN x2 / GNSS / SD

Internal connections use wires; block boundaries use local labels


CAN1 / ESP32 TWAI
U20
MCP2562FD-E/P
L20
67R@100MHz
JP1
Jumper_2_Open
J20
B4P-VH(LF)(SN)
D50
PESD1CAN-AU
CAN1_TX
CAN1_RX
CAN1_STBY
CAN1H
CAN1L
CAN1_TERM
CAN1_AUX5V
CAN2 / MCP25625 SPI
U21
MCP25625T-E/SS
L21
67R@100MHz
JP2
Jumper_2_Open
J21
B4P-VH(LF)(SN)
D51
PESD1CAN-AU
Y1
16MHz
CAN2_SCK
CAN2_MOSI
CAN2_MISO
CAN2_CS_N
CAN2_STBY
CAN2_INT_N
CAN2_RESET_N
CAN2_OSC1
CAN2_OSC2
CAN2H
CAN2L
CAN2_TERM
CAN2_AUX5V
GNSS / ACTIVE ANTENNA
Q30
2N7002K
U30
Teseo-LIV3FL
L30
47nH
C70
47pF
J30
GNSS_RESET_CTRL
GNSS_RESET_N
GNSS_RXD
GNSS_TXD
GNSS_PPS
GNSS_RF_IN
GNSS_ANT
GNSS_ANT_BIAS
I2C_SDA
I2C_SCL
microSD / SPI MODE
J31
DM3AT-SF-PEJM5
SD_CLK_CARD
SD_CLK
SD_MOSI_CARD
SD_MOSI
SD_CS_CARD_N
SD_CS_N
SD_MISO_CARD
SD_MISO
SD_DAT2
SD_DAT1
SD_DET_N
+5V
+3V3
GND

External active antenna via SMA; assign J30 footprint later; 50 ohm RF trace

DESIGN REFERENCES (page / figure)
[REF-MCU] ESP32-S3-WROOM-1/-1U DS p.10-12, p.41 Fig.9-1
https://www.espressif.com/sites/default/files/documentation/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf
[REF-USB] ESP32-S3 HW Guide p.18-19 Fig.10; official USB-OTG schematic sheet 4
https://docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/esp32s3/esp-hardware-design-guidelines-en-master-esp32s3.pdf
https://dl.espressif.com/dl/schematics/SCH_ESP32-S3_USB_OTG.pdf
[REF-CAN1] MCP2562FD DS p.6 Fig.1-2
https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP2561-2FD-High-Speed-CAN-Flexible-Data-Rate-Transceiver-DS20005284.pdf
[REF-CAN2] MCP25625 DS p.5 Fig.1-2, p.28-29 crystal circuit
https://ww1.microchip.com/downloads/en/DeviceDoc/MCP25625-CAN-Controller-Data-Sheet-20005282C.pdf

DESIGN REFERENCES / PROCUREMENT
[REF-GNSS] Teseo-LIV3FL UM2980 p.3 Fig.2, p.9-10 active antenna, p.12 Fig.11
https://www.st.com/resource/en/user_manual/um2980-hardware-specifications-for-teseoliv3fl-stmicroelectronics.pdf
[REF-SD] Hirose DM3 catalog p.3 + Espressif SD pull-up requirements
https://www.hirose.com/en/product/document?documentid=D49662_en&documenttype=Catalog&lang=en&series=DM3
https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/api-reference/peripherals/sd_pullup_requirements.html
[REF-PWR] LDR6321 Fig.4 / TCKE800NL app+land / OKL-T/6-W12 / LM2940 / XC8107
https://akizukidenshi.com/goodsaffix/ldr6321.pdf https://akizukidenshi.com/goodsaffix/TCKE800NL.pdf
https://akizukidenshi.com/goodsaffix/OKL_T6_W12.pdf https://akizukidenshi.com/goodsaffix/lm2940.pdf
https://akizukidenshi.com/goodsaffix/xc8107.pdf

Akizuki availability checked 2026-08-14. Power: LDR6321 117569 / TCKE800NL 116580 / OKL-T/6-W12N-C 106187 / NJM78L05UA 104699 / USB-C 114356. CAN2 keeps MCP25625 because MCP2515-I/P also requires a separate transceiver.

IC decoupling capacitors and functional RC, crystal and VBUS capacitors are shown
USB-C J2 is a dedicated OTG host/source port; board power enters only at J1
All specified purchasable parts: Akizuki Denshi, checked 2026-08-14

Sheet: /
File: logger_v1.kicad_sch

Title: RaceCapture Compatible Data Logger / ESP32-S3

Size: A1	Date: 2026-08-14	**Rev: A / v1**
KiCad E.D.A. 9.0.6		Id: 1/1